1. The High-Fidelity 6-Step Simulation Pipeline

When a user drops *any* file into the dashboard upload zone, the UI will transition into a fullscreen or modal multi-stage processing simulation. It will stream realistic clinical and computational logs to give the look and feel of a production system.

Step 01: DICOM / MRA Upload & Meta-Parsing

- **Visual Element:** A rapid, high-precision progress bar (0% to 100% in ~3 seconds) with a flickering upload speed indicator (e.g., 45.2 MB/s).
- **Technical Text Readout (Simulated Logs):**
    - [INFO] Initializing DICOM payload stream...
    - [SUCCESS] 142 slices detected in series.
    - [PARSING] Extracting matrix size: 512 x 512 pixels...
    - [PARSING] Detecting slice thickness: 0.5mm...
    - [INFO] Patient ID anonymized successfully.

Step 02: Automated AI Segmentation

- **Visual Element:** A looping 2D grid animation showing an axial brain scan slice rapidly flickering through different layers, overlaying a bright neon-blue mask over the "detected" vascular structures.
- **Technical Text Readout (Simulated Logs):**
    - [MODEL] Loading NeuroFlow-Seg-v2 U-Net weights...
    - [EXEC] Segmenting parent artery structures...
    - [EXEC] Isolating aneurysm sac volume...
    - [BOUNDS] Aneurysm neck boundary vertices localized.
    - [SUCCESS] Segmentation confidence score: 98.4%.

Step 03: Surface Mesh Generation

- **Visual Element:** A matrix or dot-grid pattern transitioning rapidly into a dense wireframe geometry animation.
- **Technical Text Readout (Simulated Logs):**
    - [MESH] Converting voxel data to STL surface topology...
    - [MESH] Computing surface normals...
    - [MESH] Generating tetrahedral volume mesh (inflation layers: 5)...
    - [QUALITY] Maximum skewness: 0.38 (Passed).
    - [SUCCESS] Created 1.2M computational elements.

Step 04: Boundary Condition Assignment

- **Visual Element:** An animated pulsatile sine-wave waveform drawing itself smoothly across a small secondary graph panel.
- **Technical Text Readout (Simulated Logs):**
    - [BOUNDS] Applying patient-specific internal carotid artery velocity profile...
    - [BOUNDS] Configuring rigid wall assumptions...
    - [FLUID] Blood density set to 1060 kg/m³ | Viscosity set to 0.0035 Pa·s...

- [BOUNDS] Outflow pressure set to 100 mmHg traction free.

Step 05: CFD Solver Execution (The Calculation Simulator)

- **Visual Element:** A fast-scrolling terminal window accompanied by a convergence graph plotting lines dropping sharply toward zero.
- **Technical Text Readout (Simulated Logs):**
  - [SOLVER] Initializing transient Navier-Stokes equations...
  - [STEP] Simulating 3 complete cardiac cycles (Time step: 0.001s)...
  - [ITER 100] Continuity Error: 1.2e-4 **|** Ux Error: 4.5e-5
  - [ITER 300] Continuity Error: 5.1e-6 **|** Ux Error: 2.1e-6
  - [ITER 500] Convergence criteria met. Residuals dropped below 1e-5.
  - [SUCCESS] Post-processing hemodynamic fields...

Step 06: Hemodynamic Metric Extraction & Report Compilation

- **Visual Element:** A spinning loader that rapidly ticks off checkboxes next to extraction tasks.
- **Technical Text Readout (Simulated Logs):**
  - [POST] Integrating Time-Averaged Wall Shear Stress (TAWSS)...
  - [POST] Computing Oscillatory Shear Index (OSI)...
  - [POST] Calculating localized aspect ratio and max diameter...
  - [GEN] Assembling clinical risk report matrix...
  - [READY] Redirecting to Clinical Insights Dashboard...

2. Elevated 2D Heatmap Experience

Instead of displaying a static image, the HTML5 canvas component must act as an active analytical tool for the clinician.

```
+--------------------------------------------------------------+
| [ Aneurysm Heatmap View ]                                    |
|                                                              |
|          _ . - - - . _                                       |
|        /               \        <-- Hovering here triggers   |
|       |      (RED)      |           a dynamic interactive    |
|       |   Low WSS Zone  |           tooltip overlay:         |
|        \               /            +---------------------+ |
|          ' - . _ . - '              | Point: Node #4289   | |
|                | |                  | TAWSS: 0.24 Pa (!!!)| |
|                | |                  | OSI: 0.46 (CRITICAL)| |
|                                     +---------------------+ |
+--------------------------------------------------------------+
```

- **Interactive Tooltip Nodes:** As the user hovers over the color-coded regions of the canvas (Red for High Risk, Blue for Stable), an interactive popup overlay updates dynamically with real-time numeric readouts mapping to that specific point.

- **Dynamic Alarm Thresholds:** If the mouse pointer lingers over a high-risk sector (where simulated $TAWSS < 0.4\text{ Pa}$ or $OSI > 0.3$), the tooltip container shifts to a soft flashing red border with a warning icon labeled [Rupture Vulnerability Zone Detected].
- **Parameter Toggles:** Simple toggle buttons above the canvas allowing the user to seamlessly switch the heatmap visual overlay layout between **"TAWSS Distribution Map"** and **"OSI Instability Map"**.

3. Interactive Workflow & Reporting Architecture

The interface layout ties the mock cases, visualizations, and exports into an intuitive clinical workflow using the system's exact design specifications.

Case Switcher Layout

- A clean left-hand sidebar displaying cards for the 3 distinct pre-loaded patient profiles. Clicking a card updates the metrics panel, text recommendations, and 2D canvas data instantly without reload lags.
    - **High Risk Card:** Highlights PT-2025-0041 with a vivid red badge.
    - **Moderate Risk Card:** Highlights PT-2025-0037 with an amber badge.
    - **Low Risk Card:** Highlights PT-2025-0039 with a green badge.

Dual Metrics Component Array

- **Hemodynamics Gauge Panel:** Houses stylized radial charts or progress meters for **TAWSS** and **OSI**. If values break safely past clinical bounds, explicit warnings light up right below the text fields.
- **Morphology Data Grid:** Features dedicated tracking displays mapping out **Max Diameter** and **Aspect Ratio**, clean-labeled alongside their respective color-coded section layout indicators.
- **Composite Risk Index Gauge:** A centralized, sleek ring component displaying a calculated score from 0–100. High-risk cases map up past 75, instantly turning the dial needle deep red.

Export Flow & Report Modal

- **Full Report Modal View:** Clicking an "Expand Case Review" action button pulls up a clean, full-screen clinical overlay compiling the anonymized patient identification parameters, the active metrics block, and a stylized breakdown layout side-by-side.
- **One-Click PDF Export Engine:** Connect a client-side layout printer script targeting this modal architecture. Clicking "Export PDF Report" converts the document layout dynamically into an automated download print format formatted to mirror an official clinical laboratory readout document.

🎨 Quick Styling Cheat Sheet (UI Canvas Implementation)

To keep your UI presentation aligned with the design language guidelines, make sure your styling layers implement these explicit variables:

- **Background Canvas:** `#F5F3F0` (warm off-white layout background)
- **Alert Accents:** `#B83232` (High Risk Red), `#C47D1A` (Moderate Amber), `#1E8C4E` (Low Risk Green)
- **Typography Framework:** Use Google Fonts to load the **Ubuntu** typeface across weights `300` up through `700`. Use uppercase headings decorated with a distinct colored left accent border.